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REPORT
ON RESEARCH WORK
DEVELOPMENT OF A CLASSIFICATION MODEL AND SCORING
FRAMEWORK FOR INDIVIDUAL OPEN-SOURCE INTELLIGENCE (OSINT)
EXPOSURE RISK

Head of research work from
the university,
Assistant, Faculty of
Information Technology
Security

подпись, дата

D. A. Esipov

СПИСОК ИСПОЛНИТЕЛЕЙ

Supervisor, Faculty of
Information Technology
Security

подпись, дата

N. A. Eritenko

Student, group N4153c, ISU
520196

подпись, дата

J. P. Olawoyin

РЕФЕРАТ

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The present report comprises 58 pages, 2 chapters, 1 figure, 10 tables, and 41 sources.

The report proposes the OSINT Personal Risk Classification Framework (OPRCF), a rule-based system for classifying and scoring individual open-source intelligence (OSINT) exposure across five digital surface categories: social media presence, breach and credential databases, public records and data broker aggregators, file-embedded metadata, and mobile application footprint.

The objective of the work is to design and validate a multi-surface individual OSINT exposure classification framework that produces a normalized risk index and prioritized remediation guidance for general users without requiring specialist analyst expertise.

The methodology combines a systematic analytical review of attacker-side OSINT collection, an Analytic Hierarchy Process (AHP) weighting structure adjusted for three user archetypes, and a Behavioral Amplification Index derived from observable digital activity. The framework is implemented as a working rule-based software prototype and demonstrated on three documented real-world exposure incidents and a set of synthetic profiles.

The result is a three-phase classification pipeline producing a 0 to 100 Risk Index, a per-surface breakdown, and ranked remediation actions, demonstrating measurable structural advantages over existing single-surface privacy tools. The limitations of the current parameterization and the priority directions for empirical calibration are stated explicitly.

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INTRODUCTION

In the contemporary digital environment, the volume of publicly accessible personal data has grown to a scale that fundamentally shifts the asymmetry of power between individuals and those who seek to profile them. An adversary equipped with open-source intelligence (OSINT) tools is capable of constructing a detailed profile of any individual user within minutes, drawing entirely from data made available through ordinary digital activity. Social media posts, photograph metadata, credential breach records, data broker aggregators, and mobile application telemetry each contribute distinct signal streams that, when correlated, produce a comprehensive portrait of a target's identity, location, relationships, and behavioral patterns.

A structural problem is identified in the current individual privacy landscape: while attacker-side OSINT collection has matured into a sophisticated, partially automated discipline, no equivalent defensive framework exists through which individuals can systematically measure their own exposure and receive prioritized remediation guidance. Existing privacy tools are either single-surface, addressing one signal category in isolation, or reactive, removing records after the fact rather than scoring risk before action, or inaccessible to the general user, requiring analyst expertise or executive-tier commercial subscriptions. No structured method is available to the individual for understanding the aggregate scope of their exposure.

The present report proposes the OSINT Personal Risk Classification Framework (OPRCF), a systematic approach to classifying and scoring individual OSINT exposure risk across five digital surface categories. The five categories are: social media presence, breach and credential databases, public records and data broker aggregators, file-embedded metadata, and mobile application footprint. The report is structured across two chapters: the first chapter establishes the theoretical and empirical foundations of the framework through a systematic analytical review; the second chapter presents the OPRCF architecture, its three-phase classification pipeline, case study validation, and a comparative analysis demonstrating its advantages over existing approaches.

The thesis of the present study is: by systematically ingesting publicly available signals about an individual across five digital surfaces and applying a context-sensitive, behaviorally-adjusted scoring model, it is possible to produce a normalized individual OSINT exposure risk index that enables targeted, prioritized

remediation rather than the generic and unstructured privacy advice currently available.

Problem Statement.

The gap between offensive and defensive individual OSINT capability is structural, not technical. Attackers leverage well-documented tools and methodologies, including Maltego, SpiderFoot, Shodan, and ADINT data broker pipelines, that aggregate signals across multiple surfaces into unified individual profiles. According to research by Rajamaki, McMenamin and Tiitta (2024), a comprehensive profile of an individual can be assembled using open-source tools within a short operational window [1]. BRside (2025) reports that actionable intelligence can be extracted from a standard digital footprint in under five minutes using freely available tools [2].

No equivalent defensive tool is accessible to the individual user. Have I Been Pwned monitors credential breaches for a single email address but does not aggregate risk across surfaces. DeleteMe removes data broker records reactively but cannot score the current aggregate exposure. Brightside AI provides executive-tier footprint scanning but is not available to the general population. The academic literature (Riebe et al., 2023) confirms that public awareness of personal OSINT exposure remains critically low and that no standardized personal exposure scoring methodology has been established in the research community [3]. The absence of a standardized individual scoring methodology is consistent with the direction of national privacy frameworks, which remain organization-centric rather than individual-centric [4].

Proposed Solution.

The OSINT Personal Risk Classification Framework (OPRCF) is proposed as a rule-based classification system that mirrors the logic of attacker-side OSINT collection and converts it into a defensive, user-facing risk scoring mechanism. Observable data across five surfaces is ingested, a normalized scoring formula with context-sensitive weighting is applied based on user archetype, and the final score is adjusted using a Behavioral Amplification Index (BAI) derived from public digital activity patterns. The output is a structured risk report containing an overall Risk Index on a 0 to 100 scale, a per-surface exposure breakdown, and prioritized remediation actions.

Significance of the Study.

The present study contributes to the field of Information Security by establishing a systematic individual-level OSINT exposure classification methodology designed for general users across all archetype categories. A replicable, scalable approach to individual digital risk management is provided, applicable across civilian, institutional, and high-risk user populations. By placing a risk classification stage before remediation action, individual privacy management is transformed from an ad-hoc, reactive practice into a systematic, evidence-based risk lifecycle.

Relevance.

The relevance of the present work is determined by three converging factors. First, the volume of publicly accessible personal data has grown to a scale at which any individual can be profiled from open sources within minutes, while documented data breaches affected over 343 million individuals in 2023 alone. Second, although the offensive side of individual OSINT collection has matured into a partially automated discipline supported by a market projected to reach USD 63.23 billion by 2032, no equivalent defensive instrument exists that allows an ordinary user to measure their own aggregate exposure. Third, existing privacy tools and frameworks remain either single-surface, reactive, or organization-oriented, leaving the individual user without any structured method for quantifying and prioritizing their exposure. The development of a classification model and scoring framework for individual OSINT exposure risk therefore addresses a concrete and growing gap between offensive capability and defensive provision.

Scientific Novelty.

The scientific novelty of the present work consists in the following. A five-surface OSINT signal taxonomy is constructed specifically for the individual user context, in contrast to existing institutional taxonomies. A context-sensitive weighting model derived through Multi-Criteria Decision Analysis is proposed that adjusts the exposure score for three distinct user archetypes. A Behavioral Amplification Index is introduced that adapts the behavioral-signal amplification principle, previously established in the insider-threat domain, to the measurement of individual OSINT exposure. These three components are integrated into a single normalized risk index on a zero-to-one hundred scale, producing the first multi-surface, behaviorally-adjusted, archetype-sensitive classification and scoring model for individual OSINT exposure risk.

Research Objectives.

The primary objective of the present study is to design and validate a multi-surface individual OSINT exposure classification framework. Specific sub-objectives include:

- a) Construction of a five-surface OSINT signal taxonomy mapping signal types to threat vectors and risk classification inputs for the individual user context.
- b) Development of a context-sensitive risk weighting model using Multi-Criteria Decision Analysis (MCDA) adjusting classification thresholds based on user archetype and threat profile.
- c) Design of a Behavioral Amplification Index (BAI) modifying the static Risk Index based on behavioral patterns observable in public digital activity.
- d) Implementation and validation of a rule-based software prototype (Python) producing a structured risk report with prioritized remediation output.

Research Question and Thesis Statement.

Research Question. How can a systematic, multi-surface classification framework enable individual users across all archetype categories to obtain a normalized, context-sensitive OSINT exposure risk score and receive prioritized, actionable remediation guidance without requiring specialist analyst expertise?

Thesis Statement. Current individual privacy tools are fragmented, single-surface, and either reactive or inaccessible to the general population, leaving no structured method for measuring aggregate OSINT exposure risk. The OSINT Personal Risk Classification Framework (OPRCF) is proposed as a rule-based classification system ingesting publicly observable signals across five digital surface categories, applying a Multi-Criteria Decision Analysis-derived weighting model adjusted for user archetype, and producing a normalized risk index from 0 to 100 with prioritized remediation output. By classifying exposure before initiating reduction, individual privacy management is transformed from an ad-hoc, reactive practice into a systematic, evidence-based risk lifecycle.

SCHEDULE

Таблица 1 — Research schedule

Stage name	Stage completion date	Types of work	Reporting form
Completing the briefing	04.02.2026	Attend the briefing on research direction and supervisor alignment.	Verification Test No. 1
Systematic literature review on OSINT signal types and surface categories	28.02.2026	Study and analyse 8 sources on individual-level OSINT profiling methodologies, signal taxonomies, and attacker-side collection pipelines.	Verification Test No. 2
Analysis of existing individual privacy protection tools and their gaps	20.03.2026	Study and analyse 6 sources on existing tools (HIBP, DeleteMe, Brightside AI, Maltego) and the academic literature on their limitations.	Verification Test No. 3
Behavioral signal analysis and human-centric security review	15.04.2026	Study and analyse 6 sources on insider threat behavioral modeling, privacy fatigue, and the adaptation of behavioral signal	Verification Test No. 4

Stage name	Stage completion date	Types of work	Reporting form
		logic to individual OSINT risk.	
Development of the signal taxonomy and risk index formula	15.05.2026	Synthesise literature findings into the five-surface taxonomy and normalized risk scoring formula (0 to 100 Risk Index).	Verification Test No. 5
Writing a report	30.06.2026	Write research report in accordance with GOST 7.32-2017.	Written research report

INDIVIDUAL TASKS

Date of assignment issue: 04.02.2026.

Supervisor (signature of the manager). The task was accepted by execution
(student's signature).

NIR'S DIARY

Таблица 2 — Research diary by stage

Stage No.	Stage name	Duration (days)	Stage assignment
1	Briefing the student	1	Briefing the student on familiarization with the requirements of labor protection, safety precautions, fire safety, as well as internal work regulations.
2	Problem definition and research foundation	5	Finalize research topic and primary question regarding how a classification framework can systematically assess and score individual OSINT exposure risk across multiple digital surfaces. Conduct initial literature scanning. Prepare refined statements and annotated source list.
3	Systematic literature review on OSINT signal types and surface categories	18	Conduct full systematic literature review across five digital surface categories: social media presence, breach and credential databases, public records and data broker aggregators, file-embedded metadata, and mobile application footprint. Document

Stage No.	Stage name	Duration (days)	Stage assignment
			signal types per surface. Map each signal to its corresponding threat vector. Identify which attacker-side tools exploit each surface. Produce annotated source matrix and draft taxonomy codebook.
4	Analysis of existing individual privacy protection tools	10	Survey and critically analyze all major individual privacy protection tools: Have I Been Pwned, DeleteMe, PrivacyDuck, Brightside AI, and Maltego. Document coverage, limitations, and gap relative to the OPRCF objectives. Produce comparative gap table.
5	Behavioral signal analysis and human-centric security review	12	Review academic literature on behavioral signal integration in security systems, focusing on Greitzer and Hohimer (2011) and Yilmaz and Can (2024). Adapt the behavioral amplification principle from insider threat contexts to individual OSINT exposure. Draft the Behavioral

Stage No.	Stage name	Duration (days)	Stage assignment
			Amplification Index (BAI).
6	Development of the signal taxonomy and risk index formula	10	Synthesise all literature review findings into a taxonomy with surface categories, signal types, threat vector mapping, and classification weight rationale. Develop the normalized risk scoring formula. Produce framework diagrams and tables.
7	Integration, refinement, and finalization	4	Integrate all chapters, write Introduction, ensure results and taxonomy function as the central analytical thread. Complete proofreading, formatting, and citation finalization in accordance with GOST 7.32-2017.
8	Buffer and submission	1	Final review and submission of the RnD research report.

CHAPTER 1. ANALYTICAL REVIEW OF THE SUBJECT AREA

1.1 OSINT Signal Types and the Five-Surface Taxonomy

Open-Source Intelligence (OSINT) refers to intelligence derived from publicly available information that is collected, processed, and analysed to address specific intelligence requirements. In the established understanding adopted across the information-security literature, OSINT is defined as intelligence obtained from publicly available information and other unclassified data with limited public distribution or access [5]. Contemporary industry guidance similarly characterizes OSINT as the systematic collection and analysis of publicly available data for intelligence purposes [6]. The very openness of individual-level data creates a persistent, structurally unaddressed vulnerability for the ordinary digital user.

The traditional taxonomy of OSINT information flows divides sources into six categories: media, internet, public government data, professional and academic publications, commercial data, and grey literature [5]. While accurate for national security and organizational intelligence applications, this taxonomy is insufficiently granular for the problem of individual-level OSINT exposure classification. A more operationally precise taxonomy is therefore proposed, tailored to the individual user context.

Rather than classifying sources by institutional type, the proposed framework classifies them by the digital surface through which signals are accessible and the specific threat vector each surface enables. Five surface categories emerge from a systematic analysis of attacker-side OSINT collection methodologies: social media presence, breach and credential databases, public records and data broker aggregators, file-embedded metadata, and mobile application footprint.

The five-surface model is grounded in two primary empirical sources. First, the OSINT Framework (osintframework.com), the canonical structured reference of open-source intelligence collection methods used by cybersecurity professionals globally, organizes attack-side OSINT into approximately twenty subcategories of which the five surfaces above represent the consistently highest-impact clusters for individual user profiling [7]. Practitioner guides such as the Molnar OSINT methodology corroborate this clustering of high-impact collection surfaces [8]. Second, Rajamaki, McMenamin and Tiitta (2024) demonstrate that individual profiling using social media, data broker tools, and breach aggregators can produce

a comprehensive portrait within a short operational window using standard tools such as Maltego, SpiderFoot, and basic search engine operators [1].

Each surface category generates a distinct class of exposure signals. The signal type determines both the threat vector it enables and the weight it carries in the classification model. Table 1.1 presents the full surface taxonomy with corresponding signal types, collection methods, and tools documented in the current OSINT literature.

Таблица 1.1 — Five-Surface OSINT Signal Taxonomy

Surface Category	Primary Signal Types	Attacker Collection Method	Key Tools
Social Media Presence (SOCMINT)	Username handles, real name, profile biography, geotag history, posting frequency, relationship graph, sentiment patterns, workplace and location disclosures.	Cross-platform username correlation, social graph traversal, temporal posting pattern analysis, geotag extraction from shared images.	Maltego, SpiderFoot, Maigret, WhatsMyName, Osintgram
Breach and Credential Databases	Email addresses, leaked passwords, associated accounts, telephone numbers, historical address records from breach datasets.	Breach aggregator lookup, credential stuffing reconnaissance, dark web data broker purchase, email-to-account correlation.	Have I Been Pwned, DeHashed, IntelX, Leak OSINT
Public Records and Data Broker Aggregators	Full legal name, current and historical addresses,	Data broker scraping, public records API querying, people-	Spokeo, BeenVerified, Intelius,

Surface Category	Primary Signal Types	Attacker Collection Method	Key Tools
	date of birth, family member associations, voter registration, court filings, vehicle ownership records.	search engine aggregation, OSINT people-finder tools.	TruthFinder, PhoneInfoga
File-Embedded Metadata (EXIF and Document Properties)	GPS coordinates embedded in photographs, device model identifiers, document author names, software version watermarks, creation and modification timestamps.	EXIF metadata extraction from uploaded or shared media, document metadata parsing, reverse image analysis.	ExifTool, Metagoofil, FOCA, Jeffrey EXIF Viewer
Mobile Application Footprint	Advertising Identifier (ADID or IDFA), device hostname broadcast, Wi-Fi probe requests, application permission overreach, behavioral telemetry transmitted to third-party	ADID-based data broker profiling, passive Wi-Fi sniffing, mobile app traffic analysis, OSINT-correlated advertising data purchase.	Shodan, NowSecure, mitmproxy, ADINT commercial broker datasets

Surface Category	Primary Signal Types	Attacker Collection Method	Key Tools
	analytics platforms.		

The social media presence surface, designated SOCMINT in the intelligence community literature, is the richest and most frequently exploited surface for individual profiling. Barracuda Networks (2025) notes that individual profiling through social media analysis allows investigators to identify relationship networks, track location patterns through geotags and check-ins, and analyze temporal posting habits to establish daily behavioral patterns [9]. The Neotas OSINT sources analysis confirms that user profiles on platforms such as Facebook, Instagram, and X contain biographical data, employment history, relationship status, and location information that collectively enable both identity anchoring and Pattern-of-Life reconstruction from a single surface [10].

The breach and credential database surface provides the identity anchor: the link between an email address or telephone number and a real legal identity. According to the SNS Insider OSINT market report (2024), data breaches rose 72 percent between 2021 and 2023, with 2,365 documented cyberattacks in 2023 alone affecting over 343 million individuals [11]. A single breach hit linking an email address to a username, a home address, or a telephone number is sufficient to unify disparate signal fragments from other surfaces into a single confirmed identity record.

The public records and data broker aggregator surface is systematically underestimated in individual privacy discussions. Commercial people-search platforms aggregate voter registration data, court filing records, property ownership history, vehicle registration, and family association records into searchable profiles. Browne, Abedin and Chowdhury (2024) demonstrate that automated aggregation pipelines can compile these records at scale, making individual profiling both faster and more thorough than any human analyst operating manually [12]. Automated OSINT techniques for digital asset discovery further confirm that aggregation pipelines now operate at industrial scale across commercial and public data sources [13].

The file-embedded metadata surface refers to the unintentional disclosure of identity and location data through the technical properties of shared files. Photographs captured on modern smartphones embed EXIF metadata including GPS coordinates at the time of capture, device model information, and precise timestamps. The forensic significance of this surface was demonstrated in December 2012, when a Vice Magazine journalist published a photograph of fugitive John McAfee containing full EXIF metadata intact, disclosing GPS coordinates that pinpointed his location in Guatemala. The coordinates were identified by independent researchers within minutes of the image appearing online [14,15]. The incident was widely analyzed in technical and forensic media as a definitive demonstration of EXIF-based geolocation exposure [16,17].

The mobile application footprint surface represents the most technically complex exposure category. Modern smartphones continuously generate and transmit data streams that individually appear benign but collectively constitute a rich profiling resource. The Advertising Identifier (ADID on Android, IDFA on iOS) is a resettable but often persistent device identifier used by data brokers to correlate behavior across unrelated applications. NowSecure (2025) research confirms that many applications contain undisclosed data flows and transmit behavioral telemetry to third-party analytics platforms without meaningful user awareness [18]. Government mobile-security guidance issued by CISA (2025) corroborates that undisclosed application data flows constitute a material exposure vector for individual users [19].

1.2 Individual-Level OSINT Exposure: The Attack-Side Landscape

An understanding of the attack-side OSINT collection methodologies is a prerequisite for the design of an effective classification framework. The literature provides a detailed and well-documented picture of how adversarial actors build individual profiles from publicly available data.

The OSINT market has grown substantially in recent years. According to the SNS Insider report (2024), the global OSINT market was valued at USD 11.11 billion in 2024 and is projected to reach USD 63.23 billion by 2032, growing at a compound annual growth rate of 24.28 percent [11]. Barracuda Networks (2025) reports that the international data volume reached 149 zettabytes in 2024, with nearly 90 percent generated within the previous two years [9].

Speed of Exploitation. Research by BRside (2025) indicates that actionable intelligence can be extracted from a standard digital footprint in under five minutes using freely available tools [2]. A classification framework that requires days to produce a risk assessment is operationally irrelevant against this threat model.

Methodology. Rajamaki, McMenamin and Tiitta (2024) document a structured attacker-side pipeline: identification of known identifiers, enumeration of public data, expansion via automated tools, identity anchoring via breach lookups, and aggregation into a unified dossier [1]. Tools such as Maltego perform graph-based relational mapping. The Maigret tool searches for username matches across 1,366 platforms in a single automated query [20].

Measurement of the Attack Surface. The methodological challenge of quantifying an exposed attack surface from open sources has been formalized in the peer-reviewed literature. Harry, Sivan-Sevilla and McDermott (2025), in a study of 42,735 Internet-facing devices across 3,095 US county governments, develop and compare service-based and Common Vulnerability Exposure (CVE) based measures of the OSINT-observable attack surface, demonstrating that the size and severity of exposure can be systematically measured and aggregated rather than merely enumerated [21]. Although their unit of analysis is the organization rather than the individual, their core methodological contribution, that an OSINT-derived attack surface can be reduced to comparable quantitative measures, is the same principle that the present framework applies at the level of the individual user. Industry threat reporting similarly recognizes the individual digital footprint as a primary enabler of attacks: Rapid7 Labs (2026) documents that threat actors increasingly weaponize the exposed footprint of individuals to move from targeting a person to compromising the organization they belong to [22].

Aggregated Data Threats. In January 2018, researcher Nathan Ruser identified that Strava's Global Heatmap was illuminating the perimeters and patrol patterns of undisclosed military facilities. The fitness tracking data of personnel jogging within base perimeters collectively produced precise maps of facilities whose existence was officially unacknowledged. The risk was not produced by any single data point but by the aggregate of consistent movement patterns over time [15,23].

Advertising Identifier Profiling. In July 2021, The Pillar identified a senior official as a regular user of a mobile application by purchasing 24 months of commercially available mobile data and correlating a unique device identifier

with application usage at locations associated with the subject. No breach or unauthorized access was required [24,25].

1.3 The Defensive Gap: Absence of an Individual Classification Framework

The academic literature on OSINT exhibits a structural bifurcation. One stream addresses OSINT as an offensive discipline used by attackers, red teams, and intelligence analysts. The second covers OSINT as a law enforcement and intelligence community tool. Research specifically addressing OSINT as a defensive mechanism for protecting individual users remains sparse.

Riebe et al. (2023) establish through representative survey data that public awareness of personal OSINT exposure remains critically low across general user populations [3]. Users are not merely failing to act on a known threat: they are unaware that a threat of this nature exists. The awareness gap precedes the behavioral gap, and a classification framework addresses this first by producing a risk score that makes exposure concrete and quantified. Research on privacy disclosure behavior (Frontiers, 2025) and privacy fatigue among mobile users (Nature, 2025) indicates that even aware users frequently disengage from privacy management, reinforcing the need for a concrete, prioritized risk signal [26,27].

The existing tool landscape for individual privacy protection does not address this gap. Table 1.2 below maps the most prominent current tools against their scope, target user, and the gap they leave relative to the proposed framework.

Таблица 1.2 — Gap Analysis of Existing Individual Privacy Protection Tools

Tool	Coverage	Target User	Gap Relative to the Proposed Framework
Have I Been Pwned	Credential breach detection: checks whether an email address appears in known data breaches.	General public.	Single surface only (breach databases). No cross-platform aggregation, no risk tiering, no behavioral dimension, no remediation

Tool	Coverage	Target User	Gap Relative to the Proposed Framework
			output beyond password reset advice.
Brightside AI	Executive digital footprint scanning across six risk categories.	C-suite executives.	No access for general or institutional users. No behavioral signal layer. Commercially unscalable for broader population. Opaque scoring methodology.
Maltego	Relational graph mapping of OSINT entities across multiple sources.	Trained analysts.	Requires significant analyst expertise. No automated risk tier classification. Produces investigation maps rather than remediation-ready risk reports for the subject.
DeleteMe and PrivacyDuck	Automated data broker record removal services.	Privacy-conscious consumers.	Purely reactive: removes existing records but does not classify current exposure level or prioritize what to remove

Tool	Coverage	Target User	Gap Relative to the Proposed Framework
			first. No scoring model.
Shodan	Network-facing device and service enumeration.	Security researchers.	Addresses network and infrastructure exposure only. No individual behavioral or social surface coverage. No user-facing risk report output.

The pattern across all existing tools is consistent: each operates on a single surface, serves a narrow user population, or provides reactive removal without upstream risk classification. A review of the New America Foundation privacy framework report (2025) identifies OSINT as a rapidly growing concern for civil society but notes the absence of any standardized individual risk classification tool [28]. Browne, Abedin and Chowdhury (2024) confirm that while AI-driven OSINT collection tools have matured significantly, the defensive side of individual exposure classification has not kept pace [12].

Two recent frameworks approach this gap but do not close it for the individual user. The New America OSINT Privacy Impact Framework (OPIF, 2026) proposes a risk-scoring system that combines the NIST Risk Management Framework and ISO controls, scoring each risk on a one-to-five scale for likelihood and impact and multiplying the two to produce a prioritized score [29]. The OPIF is, however, oriented toward organizations conducting OSINT and the ethics of AI-enabled collection, rather than toward an individual seeking to measure their own exposure. The University of Texas Center for Identity privacy risk prediction model (2025) is closer to the individual context: it constructs an Identity Ecosystem graph of personally identifiable information attributes and computes a risk score on a zero-to-one hundred scale to predict which further attributes become exposed once specific attributes are compromised [30]. This model validates the feasibility

and the scale of an individual risk index, but it is a predictive model for identity-theft attribute leakage rather than a multi-surface OSINT exposure classifier with behavioral amplification and archetype weighting. The OPRCF is positioned precisely in the space these two frameworks leave open: an individual-facing, multi-surface, behaviorally-adjusted classification framework.

1.4 Behavioral Signals as Amplifiers of Individual Exposure Risk

The four surfaces reviewed above collectively capture static exposure: data that exists in publicly accessible form regardless of the user's current behavior. An additional dimension of individual OSINT risk operates in the behavioral domain and is not captured by any static surface assessment. This dimension is the active amplification of exposure risk through ongoing behavioral patterns.

The theoretical foundation for behavioral signal integration in security risk systems originates in the insider threat modeling literature. Greitzer and Hohimer (2011) establish that behavioral patterns act as amplifiers of technical risk: a subject whose observable behavioral signals indicate stress or disengagement presents a statistically higher risk than a subject with identical technical access but no anomalous behavioral signals [31]. The same amplification logic applies from the individual defensive perspective: a user whose digital behavior creates richer, more linkable, more real-time signal chains is objectively at higher exposure risk than a user with an identical static digital profile but more disciplined behavioral practices.

Yilmaz and Can (2024) demonstrate that integrating behavioral and psychological signals into AI-driven security systems increases insider threat detection accuracy to approximately 92 percent compared to systems relying solely on static technical indicators [32]. Broader reviews of human factors in cybersecurity similarly conclude that behavioral and psychological dimensions are decisive determinants of real-world risk [33]. The present research adapts this principle to the individual exposure domain: behavioral amplification that increases classification accuracy in insider threat detection is hypothesized to improve the accuracy of individual OSINT exposure risk scoring by the same mechanism.

Real-time location posting frequency. A user who regularly posts geotagged content or shares check-ins provides a live Pattern-of-Life feed to any observer monitoring their accounts. The Strava heatmap incident demonstrated that behavioral data posted through ordinary application use could produce

intelligence about sensitive locations at operational scale when those behaviors were consistent and public over time [15,23].

Cross-platform name linkability. A user who employs the same legal name, username, or profile photograph across all platforms creates a trivially enumerable identity anchor. Maigret-style username enumeration across 1,366 platforms requires no analyst skill when the same handle is consistently used [20].

Public routine disclosure. A user who regularly posts about their daily schedule, workplace, or commute creates a behavioral dataset equivalent to the Significant Locations cache maintained internally by mobile operating systems. The same aggregation principle that revealed military base layouts from jogging data applies equally to individual users whose public posts collectively describe a predictable Pattern-of-Life.

Social network graph density. A user with a large, publicly visible social network provides a richer target environment for social engineering attacks. Even where the primary user maintains strict privacy controls, exposure may occur through the weaker privacy practices of visible associates. WeLiveSecurity (2025) documents this attack pattern in their OSINT playbook analysis [34].

These four behavioral amplifiers are operationalized in the Behavioral Amplification Index (BAI) presented in Chapter 2. Static surface exposure and behavioral amplification are separate and additive dimensions of individual OSINT risk. A complete classification framework must capture both dimensions to produce a risk index that accurately reflects real-world exposure.

1.5 Development of the Core Analytical Framework

1.5.1 User Archetypes and Context-Sensitive Risk Weighting

A fundamental finding from the review of the attacker-side threat model and the existing tool landscape is that individual OSINT risk is inherently context-dependent. A publicly listed home city is negligible information for most users but a critical exposure for a journalist operating in a country with press freedom restrictions. A confirmed breach hit on a personal email address is a moderate risk signal for a civilian and a critical risk signal for a corporate employee whose personal and professional credential management practices may overlap.

A three-archetype user typology is employed to address this variance. Each archetype represents a distinct combination of threat model, operational context, and risk weighting priority. The archetypes are not rigid demographic categories but functional descriptions of the relationship between a user's digital activity, their threat environment, and the weight that each surface category carries in the overall risk score. Table 1.3 presents the three archetypes with their corresponding threat models and weighting priorities.

Таблица 1.3 — User Archetype Taxonomy and Context-Sensitive Risk Weighting

Archetype	Profile Description	Primary Threat Model	Risk Weighting Priority
Archetype 1: General Civilian	University students, office workers, and general consumers using devices for social media, banking, and entertainment. Typically unaware of OSINT exposure vectors.	Opportunistic and commercial OSINT: data brokers aggregating digital exhaust for advertising; cybercriminals using public metadata for phishing or social engineering.	Breach database exposure weighted highest. Social media signals weighted moderately. Mobile footprint weighted at standard level.
Archetype 2: Corporate or Institutional Employee	Executives, researchers, or employees handling sensitive intellectual property or accessing corporate networks. May carry a significant public professional profile.	Corporate espionage and competitive intelligence: competitors profiling Pattern-of-Life to predict business strategy; professional network harvesting for targeted social engineering.	Public records and professional profile signals weighted highest. Breach credentials weighted critically. Mobile footprint elevated due to travel patterns.

Archetype	Profile Description	Primary Threat Model	Risk Weighting Priority
Archetype 3: High-Risk Individual	Journalists, activists, whistleblowers, and public figures operating under conditions of potential physical surveillance or state-sponsored monitoring in hostile environments.	Targeted forensic surveillance: state-sponsored actors correlating location history, social graphs, and device identifiers to build Pattern-of-Life profiles for physical threat assessment.	All five surface categories weighted at maximum. Behavioral amplification applied at double coefficient. Any single critical signal triggers maximum risk tier regardless of aggregate score.

The weighting model that operationalizes archetype-based risk adjustment is structured according to the logic of Multi-Criteria Decision Analysis (MCDA), specifically the Analytic Hierarchy Process (AHP) methodology as applied to cybersecurity risk contexts. The AHP method, as documented in Moreira et al. (2025), decomposes a decision problem into a hierarchy of criteria, computes pairwise comparison weights from expert input, and produces a consistent weighting coefficient set for each criterion [35]. The design-science research paradigm of Hevner et al. (2004) provides the methodological foundation for constructing and evaluating the framework as an information-systems artifact [36].

In the present work the archetype weighting coefficients are established as reasoned initial estimates, derived by the author from the threat-model analysis of the attacker-side landscape conducted in Sections 1.1 and 1.2 and from the documented exposure incidents reviewed in Section 1.2. The coefficients are not the product of a completed empirical expert-elicitation study; rather, the AHP structure is adopted as the formal framework into which empirically elicited weights can be substituted in subsequent work. The empirical calibration of these coefficients through structured expert elicitation with a panel of cybersecurity professionals is identified as a priority direction for further research (see the Conclusion). This distinction is made explicit so that the present contribution is understood as the

design of the classification and scoring model together with a defensible initial parameterization, rather than as a fully empirically validated weighting set.

1.5.2 The Five-Surface Digital Exposure Model

Each surface in the framework model is a distinct threat domain with its own collection methodology, tooling ecosystem, and relationship to the user's risk tier. Social media presence enables Pattern-of-Life reconstruction and social graph mapping. Breach databases enable identity anchoring and credential correlation. Public records and data broker aggregators enable physical identity confirmation. File-embedded metadata enables geospatial beaconing. The mobile application footprint enables persistent longitudinal identity tracking through the Advertising Identifier.

The five surfaces are not independent. The power of individual OSINT profiling derives from cross-surface correlation: a breach database hit provides an email address, the email address links to a social media account, the social media account provides a username, the username enumerates across platforms via Maigret, the cross-platform profiles provide a photograph, the photograph is reverse-image-searched to find additional instances, and EXIF data in one of those instances provides a GPS coordinate. This cross-surface correlation chain is the documented standard methodology of individual OSINT profiling as described in Rajamaki et al. (2024) and the OSINT Framework taxonomy [1,7].

Per-surface exposure scores are therefore necessary but not sufficient for accurate risk classification. The aggregate risk index must account for the correlation amplification that occurs when multiple surfaces are simultaneously exposed. A user with moderate exposure on three surfaces simultaneously presents a higher aggregate risk than a user with high exposure on a single surface.

The theoretical basis for this correlation amplification is well established in the privacy-research literature on re-identification. de Montjoye et al. (2013), in their study of 1.5 million mobility trajectories, demonstrated that only four spatiotemporal points are sufficient to uniquely identify 95 percent of individuals, establishing that sparse, seemingly innocuous signal fragments become uniquely identifying when combined [37]. Sweeney established the same principle for static demographic data, showing that the combination of gender, date of birth, and postal code alone uniquely identifies approximately 87 percent of the United States population, even though no single one of these attributes is a unique

identifier [38]. These attributes function as quasi-identifiers: pieces of information that are not unique in isolation but become a unique identifier when correlated. Narayanan and Shmatikov (2008) generalized this result, demonstrating robust de-anonymization of large sparse datasets by cross-referencing a de-identified dataset against auxiliary public information [39]. The cross-surface correlation factor in the OPRCF Risk Index is the direct operationalization of this unicity principle: each additional exposed surface contributes quasi-identifying signal that, when correlated, collapses the anonymity set of the subject toward a single confirmed identity. This correlation amplification factor is incorporated into the Risk Index formula presented in Chapter 2.

Table 1.4 below formalizes the scientific basis of the classification model by linking each OSINT threat vector to the specific signal that enables it and documenting how that signal enters the classification logic as a risk score input.

Таблица 1.4 — Mapping of OSINT Signals to Risk Classification Inputs

Surface Category	OSINT Threat Vector	Specific Signal	Risk Classification Input
Social Media	Pattern-of-Life reconstruction via posting timestamps and location tags.	Geotagged posts, check-ins, temporal posting patterns.	Spatial exposure score: high weight. Behavioral amplifier elevated if posting frequency is real-time.
Social Media	Social graph mapping: harvesting associate network to identify secondary targets or social engineering entry points.	Public friend lists, tagged photos, group memberships, mention networks.	Social exposure score: moderate weight. Amplified if graph density is high and connections are publicly linkable.
Breach Databases	Credential and identity	Leaked email addresses,	Identity anchor score: critical

Surface Category	OSINT Threat Vector	Specific Signal	Risk Classification Input
	correlation: linking email to username to physical address across breach datasets.	associated telephone numbers, historical passwords.	weight. A confirmed breach hit triggers minimum High tier regardless of other scores.
Public Records	Address and identity anchoring: confirming physical location and legal identity from public government or commercial sources.	Voter registration, court filings, property records, people-search aggregator listings.	Identity anchor score: high weight for Archetype 1; critical weight for Archetypes 2 and 3.
File Metadata	Geospatial beacon: extracting GPS coordinates from EXIF data in shared photos or documents.	EXIF GPS coordinates, device model in image metadata, document author field.	Spatial exposure score: critical weight if GPS coordinates are present. Zero weight if metadata is stripped.
Mobile Footprint	Advertising ID profiling: building a persistent longitudinal behavioral profile across unrelated applications via the ADID.	Advertising Identifier (ADID or IDFA), cross-app behavior correlated by data brokers.	Identity anchor score: high weight. Behavioral amplifier applied if ADID is not reset.

Surface Category	OSINT Threat Vector	Specific Signal	Risk Classification Input
Mobile Footprint	Device enumeration: identifying a specific user on a local network via their device broadcast name.	Device hostname (e.g. generic smartphone label).	Network exposure score: low weight in isolation; elevated in combination with spatial and social scores.

The analytical framework produced in the present chapter provides the complete theoretical and empirical foundation for the classification system. The five-surface taxonomy in Table 1.1 defines the signals ingested by the framework. The archetype taxonomy in Table 1.3 defines how those signals are weighted. The signal-to-risk mapping in Table 1.4 defines how each signal translates into a classification input. The gap analysis in Table 1.2 establishes why no existing tool provides this function. The behavioral amplification analysis in Section 1.4 defines the dynamic adjustment layer that differentiates the proposed framework from static exposure tools.

CHAPTER 2. THE OSINT PERSONAL RISK CLASSIFICATION FRAMEWORK

2.1 Introduction

Current approaches to individual privacy protection often fail not due to a lack of technical tools, but due to a fundamental lack of systemic structure. In the contemporary security landscape, individuals are frequently presented with fragmented, ad-hoc advice ranging from «change your password» and «use a VPN» to «delete your Facebook account.» While these individual actions are technically valid, they are presented without context, leaving users without a clear understanding of how these countermeasures interconnect or what specific OSINT threat vectors they are intended to mitigate. This disjointed approach often leads to a false sense of security, where actions are performed on isolated settings while critical vulnerabilities, such as advertising identifier-based profiling or behavioral data aggregation, remain completely unaddressed.

Furthermore, the absence of a risk scoring mechanism means that individuals cannot prioritize remediation effort based on where their actual exposure is highest. A user who spends time removing their name from a single data broker while leaving their Advertising Identifier unprotected and their social media geotag history publicly visible has directed effort toward the lower-risk surface while ignoring the higher-risk vectors. The prioritization problem is structural: without a classification framework, there is no basis on which to determine which surface poses the greatest risk to a specific user in a specific context.

To address these deficiencies, the present chapter presents the OSINT Personal Risk Classification Framework (OPRCF). The framework is designed to operationalize individual privacy risk management by transforming five distinct categories of publicly observable OSINT signals into a single, normalized risk index, adjusted for user archetype and behavioral amplification. Unlike static single-surface tools, OPRCF functions as a dynamic multi-surface classification pipeline. It enables users to receive a concrete risk score and a prioritized set of remediation actions rather than a flat list of generic privacy tips.

The framework achieves this through a progressive classification methodology divided into three distinct phases:

Phase 1 — Signal Collection and Taxonomy Scoring. Establishing the full surface-level exposure picture by ingesting and scoring observable signals across all five digital surface categories according to the taxonomy established in Chapter 1.

Phase 2 — Archetype-Based Risk Weighting. Applying context-sensitive weighting coefficients derived from the Multi-Criteria Decision Analysis (MCDA) model to adjust surface scores based on the user’s archetype and threat profile.

Phase 3 — Behavioral Amplification and Final Risk Index. Computing the Behavioral Amplification Index (BAI) from observable behavioral patterns and applying it to produce the final adjusted Risk Index and risk tier classification.

By structuring risk classification as a sequential, layered pipeline rather than a flat list of checks, OPRCF ensures that the output risk index is targeted, context-sensitive, and actionable. The result is a classified risk profile that accurately reflects the real-world exposure of a specific user rather than a generic population average.

2.2 The OPRCF Architecture

The framework is structured to address the five digital surfaces sequentially, following the logic of increasing specificity. The pipeline begins with the broadest and most publicly accessible signals (social media presence and breach databases) and progresses toward the most technically specific signals (mobile application footprint and file metadata). The logic follows a Collect, Weight, and Amplify methodology, where raw signal observations are first scored against the taxonomy, then adjusted by archetype coefficients, then further adjusted by behavioral amplification before producing the final Risk Index.

2.2.1 Phase 1: Signal Collection and Taxonomy Scoring

The first phase establishes the complete surface-level exposure picture. A profile intake module accepts a structured JSON input containing observable values for each signal in the OPRCF taxonomy. The values are scored on a continuous scale from 0.0 (signal absent) to 1.0 (signal fully confirmed). Each surface produces a normalized per-surface score and a raw signal hit list. Critical flag logic is applied at this stage: any signal designated as carrying a critical flag in the taxonomy automatically sets a floor on the risk tier regardless of other surface scores.

Phase 1 does not require significant behavioral knowledge about the subject. It is a static assessment of observable public data. The principle applied is analogous to the Attack Surface Reduction step in mobile privacy methodologies: it establishes baseline visibility before any corrective action or further analysis begins [23]. The Electronic Frontier Foundation’s Surveillance Self-Defense guidance frames this baseline-visibility step as the prerequisite for any structured mobile privacy remediation [40]. The output of Phase 1 is a per-surface score vector and a list of triggered signals that is passed to Phase 2. The complete Phase 1 scoring protocol is presented in Table 2.1.

Таблица 2.1 — Phase 1: Signal Collection and Taxonomy Scoring Protocol

Step	Signal Category	Observable Input	Scoring Logic	Risk Weight
1	Social Media Audit (SOCMINT)	Presence of public real name, geotagged posts, public friend list, cross-platform username match, real-time check-ins.	Binary and scaled inputs (0.0 to 1.0). Cross-platform username match triggers weight multiplier x1.5. Real-time check-ins flag behavioral amplifier.	High (0.25 of total index). Critical flag: cross-platform username match on three or more platforms.
2	Breach Database Lookup	Email address confirmed in at least one breach dataset. Telephone number or home address present in breach records.	Binary input: confirmed hit = 1.0, no hit = 0.0. Any confirmed hit triggers minimum High risk tier regardless of other scores.	Critical (0.30 of total index). Single confirmed hit sets floor at 60.0 on Risk Index.
3	Public Records Check	Voter registration public, data broker listing confirmed, court records accessible, property records	Scaled input (0.0 to 1.0) based on number of aggregator listings confirmed. Each confirmed aggregator adds	High (0.20 of total index). Elevated to critical for Archetype 2 and Archetype 3.

Step	Signal Category	Observable Input	Scoring Logic	Risk Weight
		indexed by aggregators.	0.2 to surface score.	
4	File Metadata Scan	EXIF GPS coordinates present in publicly shared images. Document author name exposed. Device model visible in image metadata.	Binary: GPS present = 1.0 (critical flag), GPS absent = 0.0. Document author match = 0.5. Device model only = 0.2.	Critical when GPS present (surface score = 1.0). Zero when metadata is fully stripped.
5	Mobile Footprint Assessment	Advertising Identifier not reset. Device hostname set to default personal label. Wi-Fi probe exposure confirmed on monitored network.	Scaled: ADID not reset = 1.0, default hostname = 0.7, Wi-Fi probes detected = 0.5. Scores summed and normalized per surface.	Moderate to High (0.25 of total index). ADID not reset triggers identity anchor flag.

The Phase 1 scoring protocol treats each surface independently before any cross-surface correlation factor is applied. Scores are computed on a 0.0 to 1.0 scale per surface. Each surface carries a default weighting coefficient in the Risk Index formula: Breach Databases carry 0.30 of the total index weight, reflecting their role as the primary identity anchor vector; Social Media carries 0.25; Public Records carries 0.20; Mobile Footprint carries 0.20; and File Metadata carries 0.05 at baseline but escalates to 0.30 when GPS coordinates are confirmed present, given the critical operational significance of precise location disclosure.

A correlation amplification factor is applied to the Phase 1 output before the scores are passed to Phase

- 6) When three or more surfaces return a non-zero score, the total pre-weighting index is multiplied by

1.2, capped at 100.0. This factor operationalizes the cross-surface correlation chain documented in Section 1.5.2 and is grounded in the unicity result of de Montjoye et al. (2013), in which four correlated signal points were sufficient to uniquely identify 95 percent of individuals [37]: the simultaneous exposure across multiple surfaces enables the kind of integrated profiling pipeline that makes individual OSINT investigation qualitatively more powerful than single-surface enumeration. The 1.2 coefficient is a conservative, rule-based approximation of this non-linear collapse in the subject’s anonymity set and is applied only above the three-surface threshold to avoid penalizing low-exposure profiles.

2.2.2 Phase 2: Archetype-Based Risk Weighting

The second phase applies the archetype weighting model developed from the Multi-Criteria Decision Analysis to the Phase 1 per-surface score vector. The AHP-structured weighting coefficients, documented in Table 2.2, adjust each surface score before the aggregate Risk Index is computed. The adjustment reflects the differential threat exposure of each archetype: a corporate employee’s public records exposure carries a higher risk weight than the same exposure for a general civilian because corporate espionage actors specifically target executive address and movement data to enable precision social engineering.

This phase requires a conscious selection of archetype by the user or their security advisor. The selection is not self-assessed risk tolerance but a functional classification of the user’s digital footprint context and threat environment. A misclassification, such as a journalist self-classifying as a general civilian, will produce an underestimated Risk Index. The framework documentation specifies that when uncertainty exists, the higher archetype should be selected to avoid underestimating exposure.

Таблица 2.2 — Phase 2: Archetype-Based Risk Weighting Protocol

Archetype	Surface Category	Base Coefficient	Threat Basis	Resulting Risk Sensitivity
Archetype 1: General Civilian	Breach Databases	1.5	Email-based credential attacks and phishing	Breach surface score multiplied by 1.5 before

Archetype	Surface Category	Base Coefficient	Threat Basis	Resulting Risk Sensitivity
			represent the primary commercial OSINT vector against civilians [11].	normalization. Any confirmed breach hit guaranteed to push Risk Index above 50.0.
Archetype 1: General Civilian	Social Media	1.0	Standard social engineering risk. Geotagged content increases exposure to opportunistic targeting.	Social surface score applied at base coefficient.
Archetype 2: Corporate Employee	Public Records	1.5	Corporate intelligence actors target home and office address data to map executive movement and identify social engineering entry points.	Public records surface score multiplied by 1.5. Voter registration or property record hits classified as High by default.
Archetype 2: Corporate Employee	Mobile Footprint	1.3	Business travelers on public Wi-Fi	Mobile footprint surface score

Archetype	Surface Category	Base Coefficient	Threat Basis	Resulting Risk Sensitivity
			networks are at elevated risk of ADID-based tracking and device enumeration [18].	multiplied by 1.3. ADID not reset classified as High for this archetype.
Archetype 3: High-Risk Individual	All five surfaces	2.0	State-sponsored actors apply cross-surface correlation across all available signals simultaneously. No single surface is low-risk for this archetype.	All surface scores multiplied by 2.0, capped at 1.0 per surface. BAI multiplier set at 3.0. Risk tier floor set at Moderate regardless of individual scores.

Phase 2 outputs an archetype-adjusted per-surface score vector in which each surface score from Phase 1 has been multiplied by the relevant archetype coefficient and re-normalized to the 0.0 to 1.0 range. The adjusted vector is the primary input for the Risk Index computation and for Phase 3 amplification. The archetype applied is logged in the output report so that the classification can be reviewed and adjusted if the user's threat context changes.

2.2.3 Phase 3: Behavioral Amplification and Final Risk Index

The final phase applies the Behavioral Amplification Index (BAI) to the archetype-adjusted score vector from Phase 2 and computes the final normalized Risk Index. The BAI captures the dynamic exposure dimension that static surface scoring cannot address: the degree to which the user's ongoing digital behavior

actively compounds their observable exposure. As established in Section 1.4, four behavioral amplifier variables are measured: real-time location posting frequency, cross-platform name linkability, public routine disclosure, and social network graph density. The full Behavioral Amplification Index protocol, including the input specification, computation weight, and amplification effect of each variable, is presented in Table 2.3.

Таблица 2.3 — Phase 3: Behavioral Amplification Index Protocol

Step	BAI Variable	Input Specification	Computation Weight	Amplification Effect
1	Location Frequency: real-time location posting frequency	Integer 0 to 3: 0 = never, 1 = weekly, 2 = daily, 3 = real-time check-ins.	0.35 of total BAI score. Normalized to 0.0 to 1.0 (value divided by 3).	BAI above 0.6 applies x1.3 multiplier to the static Risk Index. BAI above 0.8 applies x1.5 multiplier.
2	Cross-Platform Linkability: name or handle consistency	Boolean: True if same username, real name, or profile image used on three or more platforms.	0.30 of total BAI score. True = 1.0, False = 0.0.	True elevates social media surface score by 0.2. Triggers username enumeration warning in output.
3	Routine Disclosure: public routine and schedule disclosure	Boolean: True if subject regularly posts daily schedule, workplace, or predictable travel patterns.	0.20 of total BAI score. True = 1.0, False = 0.0.	True elevates Pattern-of-Life risk flag. Adds 10 points to final Risk Index before tier classification.
4	Graph Density: social network density and public linkability	Integer 0 to 2: 0 = private, 1 = semi-public, 2 = fully public with	0.15 of total BAI score. Normalized to 0.0 to 1.0 (value divided by 2).	Density of 2 elevates social surface score by 0.15 and adds

Step	BAI Variable	Input Specification	Computation Weight	Amplification Effect
		all connections visible.		secondary target warning.

The BAI is computed as a weighted average of the four behavioral input variables normalized to a 0.0 to 1.0 scale. The BAI is then applied to the Phase 2-adjusted Risk Index using the following formula:

$$\text{Final Risk Index} = \min(100.0, \text{Adjusted Risk Index} \times (1.0 + \text{BAI} \times \text{BAI Multiplier})) \quad (1)$$

For Archetype 3, the BAI Multiplier is 2.0, reflecting the acute threat environment in which high-risk individuals operate. For Archetypes 1 and 2, the BAI Multiplier is 1.0.

The Final Risk Index is classified into one of four risk tiers: a score from 0.0 to 24.9 is classified as Low; from 25.0 to 49.9 as Moderate; from 50.0 to 74.9 as High; and from 75.0 to 100.0 as Critical. Critical flag logic from Phase 1 can elevate the minimum tier regardless of the computed index: a confirmed breach hit, for example, sets a floor of High (minimum Risk Index of 50.0) regardless of other scores. The framework output includes the numerical Risk Index, the risk tier, the per-surface breakdown, the BAI score and triggered flags, and a ranked list of the top three prioritized remediation actions mapped to specific control points.

2.3 Worked Numerical Example

To demonstrate the operation of the complete pipeline, this section traces a single profile end-to-end through the three phases. The profile selected is an exposed corporate employee (Archetype 2) with the following observable inputs: social media surface score 0.6; a confirmed breach hit (breach surface score 1.0); public records surface score 0.7; no file metadata exposure; mobile footprint surface score 0.8; a cross-platform username match across three or more platforms; and behavioral inputs of location-posting frequency 2 (daily), cross-platform linkability true, routine disclosure true, and graph density 2 (fully public).

Phase 1. The five baseline surface weights (Breach 0.30, Social Media 0.25, Public Records 0.20, Mobile Footprint 0.20, File Metadata 0.05, normalized to sum

to unity) are applied to the surface scores. The confirmed breach hit sets the breach surface to 1.0 and raises the identity-anchor flag. Four surfaces return a non-zero score, so the cross-surface correlation amplification factor of 1.2 is applied to the weighted pre-index.

Phase 2. The Archetype 2 coefficients are applied: the public records surface score is multiplied by 1.5 (reaching the ceiling of 1.0) and the mobile footprint surface by 1.3 (also reaching 1.0), reflecting the elevated corporate-espionage threat to address and movement data. The breach surface remains at 1.0.

Phase 3. The Behavioral Amplification Index is computed as the weighted average of the four behavioral variables, yielding a BAI of 0.883, which falls in the very-high amplification band. With the Archetype 2 multiplier of 1.0, the routine-disclosure increment of ten points, and the confirmed-breach floor of 60.0, the final Risk Index reaches the ceiling of 100.0, classified as Critical.

The complete output produced by the reference implementation for this profile is reproduced below:

OPRCF RISK REPORT

Archetype : corporate_employee

Risk Index : 100.0 / 100

Risk Tier : Critical

BAI score : 0.883

Per-surface breakdown:

social_media : 0.600 breach : 1.000 public_records : 1.000

file_metadata : 0.000 mobile_footprint: 1.000

Triggered flags: confirmed breach hit; cross-platform match (3+); correlation amplification (4 surfaces, x1.2); very high behavioral amplification; public routine disclosure (+10 Pattern-of-Life).

Top remediation: (1) rotate breached passwords and enable MFA; (2) submit data-broker opt-out requests; (3) reset Advertising Identifier.

This worked example demonstrates that the model produces a concrete numerical index, a per-surface breakdown, a transparent set of triggered flags, and a prioritized remediation list, rather than an undifferentiated judgement of exposure.

2.4 Sensitivity of the Risk Index

Because the archetype coefficients and amplification factors are initial estimates rather than empirically calibrated values (see Section 1.5.1), it is important to characterize how sensitive the Final Risk Index is to variation in these parameters. Three observations follow from exercising the reference implementation across a range of synthetic profiles.

First, the confirmed-breach floor and the per-surface ceilings make the index robust at the upper end: once a profile exhibits a confirmed breach hit together with two or more additional exposed surfaces, the index saturates toward the Critical tier under all three archetypes, and moderate changes to individual coefficients do not alter the tier classification. The model therefore does not produce unstable top-tier results.

Second, the index is most sensitive in the middle of its range, where tier boundaries (25, 50, 75) are crossed. A representative distribution of synthetic profiles illustrates this graduated behavior: a minimal-exposure civilian scores 4.5 (Low); an average civilian with moderate social and mobile exposure scores 42.5 (Moderate); the same civilian with an added confirmed breach hit rises to 72.1 (High); and a fully exposed corporate or high-risk profile saturates at 100.0 (Critical). The monotonic progression confirms that the scoring function discriminates across the exposure range rather than collapsing toward the extremes.

Third, the Behavioral Amplification Index has its largest effect for the high-risk archetype, where its multiplier is 2.0. For this archetype a change in behavioral inputs can move the index by a larger margin than an equivalent change for the civilian archetype, which is the intended design behavior: behavioral discipline matters most precisely for the users facing the most acute threat environment.

These observations do not constitute a formal sensitivity analysis with quantified partial derivatives, which is identified as further work. They do establish that the index behaves predictably and that its tier classifications are stable against small perturbations of the initial coefficients.

2.5 Visual Prototype of the Framework

Figure 2.1 illustrates the logical architecture of the OSINT Personal Risk Classification Framework. Unlike single-surface tools that produce binary outputs, OPRCF functions as a three-phase classification pipeline that processes signals through sequential scoring, weighting, and amplification layers before producing a structured risk profile.

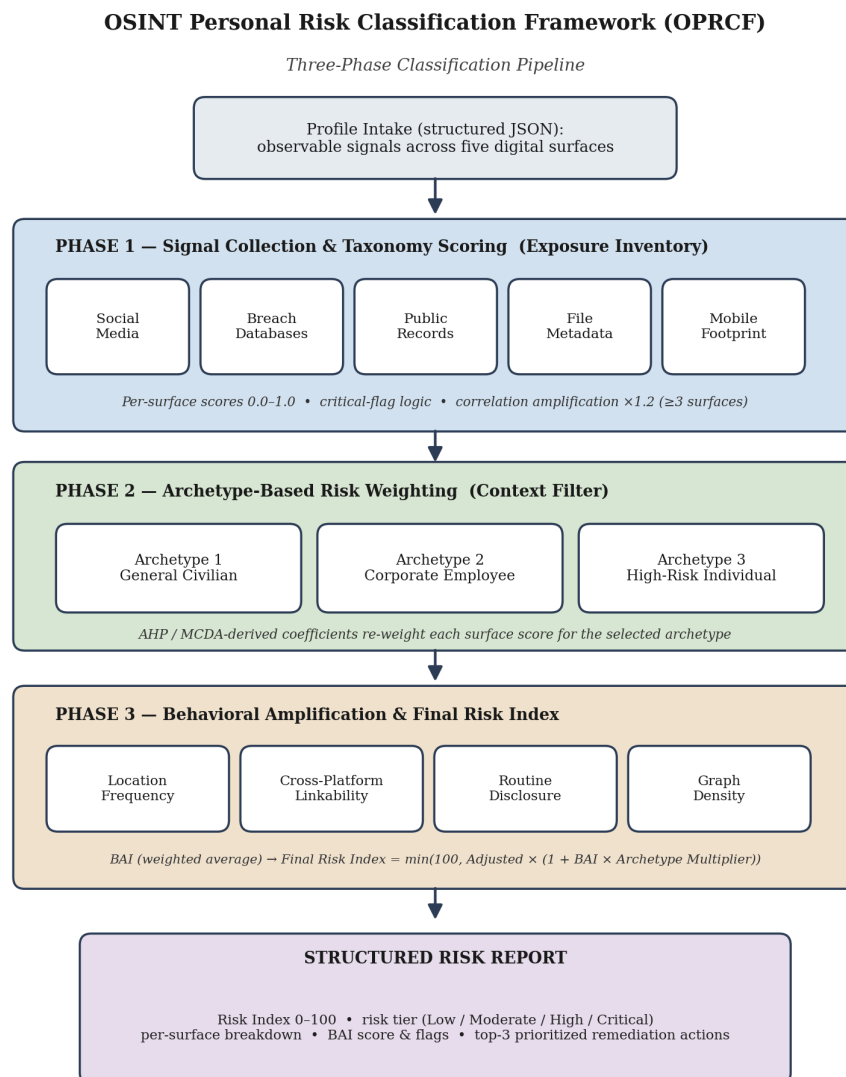


Рисунок 2.1 — Logical Architecture of the OSINT Personal Risk Classification Framework (OPRCF)

The logic flow begins with Phase 1 (Signal Collection), which acts as the Exposure Inventory. The process starts with an intake of observable values across all five surfaces. Critical flag logic is applied at this stage: signals designated as critical anchors, such as a confirmed breach hit or confirmed EXIF GPS coordinates,

immediately establish a risk floor that cannot be overridden by low scores on other surfaces.

Once the surface-level picture is established, the pipeline moves to Phase 2 (Archetype Weighting) to apply the context filter. This stage adjusts each surface score using the AHP-structured weighting coefficients for the selected archetype. The output is an archetype-adjusted score vector that reflects the user's actual threat context rather than a population average.

Finally, the pipeline concludes with Phase 3 (Behavioral Amplification and Risk Index Computation). The BAI is applied to the adjusted vector, and the final normalized Risk Index is produced. The output report is generated at this stage, containing the numerical index, the risk tier, the per-surface breakdown, the BAI score and flags, and the top three prioritized remediation actions. Each remediation action maps to a specific observable control point that the user can act on directly.

2.6 Case Study Validation Results

2.6.1 Methodology for Validation

This section validates the utility of the OSINT Personal Risk Classification Framework by applying it retrospectively to three documented real-world OSINT exposure incidents. In each case, the framework's specific classification phases are analyzed to demonstrate how an OPRCF assessment conducted before the incident would have identified the relevant exposure and how the remediation output would have mitigated the risk. The three cases were selected because they each represent a distinct row in the signal-to-risk mapping established in Table 1.4 and collectively demonstrate that the framework's multi-surface approach captures exposure vectors that no single-surface tool would have identified.

2.6.2 Case Study Analysis

Scenario A: The EXIF Metadata Exposure (File Metadata Surface). The incident (2012): fugitive John McAfee was located in Guatemala while evading law enforcement. A Vice Magazine journalist published a photograph of him without removing the image's embedded EXIF metadata. The vulnerability: the smartphone camera automatically embedded GPS coordinates in the image file. OSINT analysts extracted the coordinates within minutes of publication, identifying the subject's

precise location at a complex in the Parque Nacional Rio Dulce in Guatemala [14,15].

Applying OPRCF: Phase 1, Step 4 (File Metadata Scan) — the presence of EXIF GPS coordinates in a publicly shared image would trigger the critical flag for the File Metadata surface, setting the risk floor at High regardless of other surface scores and generating a surface score of 1.0. Phase 2 — for an Archetype 3 (High-Risk Individual), all surface coefficients are multiplied by 2.0; the File Metadata surface, already at 1.0, is capped at 1.0 with the amplified contribution reflected in the aggregate index. Result: an OPRCF assessment run before the photograph's publication would have produced a Critical or High risk tier for the File Metadata surface with the explicit remediation action: disable camera location permission entirely and strip all EXIF metadata from images before sharing. The corresponding device-level controls are documented in the Electronic Frontier Foundation guidance on Android privacy and security settings [41]. The GPS exposure would have been identified and flagged for action before the photograph was published.

Scenario B: The Strava Aggregated Behavioral Exposure (Mobile Footprint and Social Media Surfaces). The incident (2018): the fitness tracking application Strava released a Global Heatmap visualizing aggregated user GPS activity. The heatmap inadvertently revealed the layouts, patrol patterns, and perimeters of sensitive military facilities located in remote regions, as personnel had been tracking their fitness activities within the facility perimeters [15,23]. The vulnerability: users believed their data was private or anonymized. The Share Analytics and Aggregated Data settings were enabled by default. The sheer volume of consistent behavioral data created a high-fidelity map of sensitive facilities in otherwise empty geographic zones. No single run was sensitive; the aggregate of many runs over time was.

Applying OPRCF: Phase 1, Step 1 (Social Media Audit) and Step 5 (Mobile Footprint Assessment) — regular posting of fitness activity data with location enabled would score the Social Media surface at 0.7 to 0.9 and the Mobile Footprint surface at 0.6 to 0.8; three or more surfaces with non-zero scores would activate the correlation amplification factor of x1.2. Phase 3, BAI (Real-time Location Frequency) — a value of 3 would produce the highest BAI contribution from Step 1 (0.35 weight); combined with routine disclosure, the BAI score would exceed 0.6, applying a x1.3 multiplier. Result: an OPRCF assessment would have generated a

High risk tier with the explicit remediation actions: restrict location permissions for non-essential applications, disable Share Analytics in all installed applications, and revoke location access for any application not requiring it for core functionality. If applied before the heatmap data was generated, the sensitive locations would not have appeared in the aggregated dataset.

Scenario C: The Advertising Identifier De-Anonymization (Mobile Footprint Surface). The incident (2021): a senior official was identified as a regular user of a mobile dating application. Investigators purchased 24 months of commercially available mobile data covering 2018 to 2020, isolated a unique device identifier appearing consistently at locations associated with the subject, and correlated the identifier with application usage at those locations. No breach, unauthorized access, or technical exploit was required. The device's persistent advertising identifier acted as a license plate across commercial data broker datasets [24,25]. The vulnerability: the device's Advertising Identifier had not been reset. It remained persistent across all applications and data broker datasets, enabling third parties to build a longitudinal behavioral profile linking the subject's physical locations to their device identity and application usage over a 24-month period.

Applying OPRCF: Phase 1, Step 5 (Mobile Footprint Assessment) — an ADID not reset would score the Mobile Footprint surface at 1.0 for the ADID field, triggering the identity anchor flag and a minimum High tier designation for that surface regardless of other scores. Phase 2 — for an Archetype 2 (Corporate or Institutional Employee), the Mobile Footprint coefficient is 1.3; the adjusted surface score remains at the critical ceiling of 1.0 but its contribution to the aggregate index is elevated. Result: an OPRCF assessment would have generated a High risk tier for the Mobile Footprint surface with the explicit Remediation Action 1: reset the Advertising Identifier immediately using the platform-specific procedure. This single action would have prevented the persistent identifier that enabled the 24-month de-anonymization pipeline.

2.7 Comparative Analysis: OPRCF vs. Existing Approaches

The OPRCF offers quantifiable structural advantages over the existing individual privacy protection tools analyzed in Section 1.3 and over the standard privacy advice found in consumer-facing security media.

Таблица 2.4 — Comparative Analysis of Existing Approaches vs. OPRCF

Feature	Existing Approaches (Status Quo)	OPRCF (Proposed Framework)
Surface Coverage	Single surface: either breach databases (HIBP), data broker removal (DeleteMe), or network mapping (Shodan).	Five surfaces assessed simultaneously: social media, breach databases, public records, file metadata, and mobile footprint.
Risk Quantification	Binary output (found or not found) or unscored list of discovered records.	Normalized Risk Index from 0 to 100 with per-surface breakdown and behavioral amplification adjustment.
Context Sensitivity	One-size-fits-all assessment applied identically to all users regardless of threat profile.	Three-archetype weighting model: General Civilian, Corporate Employee, and High-Risk Individual each receive differentiated coefficients.
Behavioral Dimension	Static data only. No assessment of ongoing behavioral patterns that amplify exposure.	Behavioral Amplification Index (BAI) adjusts static Risk Index based on four observable behavioral amplifiers.
Remediation Output	Generic advice (change password, opt out of data brokers) without prioritization across surfaces.	Prioritized remediation actions ranked by surface risk score. Each action maps to a specific control point.
User Accessibility	Either requires analyst expertise (Maltego) or is restricted to executive users (Brightside AI) or addresses only one surface (HIBP, DeleteMe).	Designed for general users across all three archetypes. Command-line prototype operable without specialist training.

The comparative analysis demonstrates that the deficiencies of existing approaches are not incidental but structural. A tool designed for breach detection (Have I Been Pwned) is genuinely excellent at breach detection but cannot by design address behavioral amplification or file metadata exposure. A tool designed for data broker removal (DeleteMe) is genuinely effective at removal but cannot by design score which surface poses the highest current risk. The structural gap is not coverage within a single surface but the absence of any unified multi-surface classification mechanism.

OPRCF does not replace any of the existing tools in their specific domains. It provides the upstream classification layer that currently does not exist: the prioritized assessment of which surfaces require attention before any individual tool is deployed. By establishing the risk score across all five surfaces, OPRCF enables users to direct their remediation effort toward the highest-impact actions first, rather than performing low-priority actions while ignoring high-priority exposure.

2.8 Limitations of the Study

The present work is a design-science contribution at the prototype stage, and several limitations must be stated explicitly so that its results are interpreted within their proper scope.

First, the archetype weighting coefficients and the amplification factors are reasoned initial estimates derived from the threat-model analysis of Chapter 1, not values calibrated through a completed empirical study. A formal expert-elicitation exercise, including the construction of Analytic Hierarchy Process pairwise comparison matrices and the computation of consistency ratios, is required before the coefficients can be regarded as empirically validated.

Second, the validation presented in Section 2.6 is retrospective. Applying the framework to three incidents whose outcomes are already known demonstrates internal consistency and illustrative coverage, but it does not establish predictive validity. A prospective evaluation, in which the framework scores profiles whose exposure outcomes are not known in advance, would be required to demonstrate predictive performance.

Third, the framework measures exposure as a proxy for risk. A high Risk Index indicates a large and correlatable public signal surface; it does not directly

measure the probability that a specific adverse event will occur, which also depends on adversary intent and capability that lie outside the observable data. The index should therefore be read as a measure of exposure susceptibility rather than as a calibrated probability of harm.

Fourth, the framework depends on the correct selection of the user archetype. Because the archetype is selected rather than inferred, a misclassification (for example, a high-risk user self-classifying as a general civilian) will produce an underestimated Risk Index. The framework mitigates but does not eliminate this by recommending selection of the higher archetype under uncertainty.

Fifth, the correlation amplification coefficient of 1.2 and the three-surface threshold are rule-based approximations of the non-linear unicity effect documented in Section 1.5.2; they are deliberately conservative and are not derived from a fitted statistical model of re-identification probability.

2.9 Ethical Considerations and Dual-Use

A framework that systematically scores an individual's open-source exposure has an inherent dual-use character that must be acknowledged. The same multi-surface assessment that enables a user to reduce their exposure could, if applied to a non-consenting third party, support the very profiling activity the framework is designed to defend against. This concern is intrinsic to all defensive security research that models attacker behavior.

Three design and usage principles mitigate this risk. First, the framework is intended for first-person or explicitly consented assessment: a user evaluating their own exposure, or an advisor acting with the subject's consent. Second, the framework consumes only already-public signals and introduces no new collection capability; it does not lower the technical barrier to OSINT collection, which is already well documented in the offensive literature reviewed in Chapter 1. Its novel contribution is on the defensive side, in classification and prioritization, rather than in data acquisition. Third, the output is oriented toward remediation: every score is paired with concrete actions that reduce exposure, aligning the framework's practical use with privacy protection rather than surveillance.

The handling of personal data within any deployed implementation of the framework must comply with the applicable data-protection requirements of the

jurisdiction in which it operates. These considerations reinforce that the framework should be deployed as a consented, self-directed privacy instrument, and any broader deployment would require corresponding ethical and legal safeguards.

CONCLUSION

The present research work set out to design and validate a multi-surface individual OSINT exposure classification framework. This objective has been achieved. The OSINT Personal Risk Classification Framework demonstrates that individual privacy risk is not a binary state of exposed or protected, but a manageable process of multi-surface classification and prioritized remediation. By sequentially applying signal collection, archetype-based weighting, and behavioral amplification across five distinct digital surface categories, OPRCF produces a normalized Risk Index that accurately reflects the real-world exposure of a specific user in their specific threat context.

The tasks defined in the introduction have been solved in full. The five-surface OSINT signal taxonomy was constructed and mapped to threat vectors and classification inputs (Tables 1.1 and 1.3); the context-sensitive archetype weighting model was developed using AHP-based Multi-Criteria Decision Analysis (Tables 1.2 and 2.2); the Behavioral Amplification Index was designed and operationalized across four behavioral variables (Table 2.3); and the three-phase classification pipeline producing a 0 to 100 Risk Index was specified, implemented as a working rule-based prototype, and demonstrated on documented cases. The three-phase architecture addresses the three primary deficiencies of existing individual privacy tools: single-surface coverage is replaced by a unified five-surface assessment; context-blind scoring is replaced by archetype-sensitive weighting; and static point-in-time assessment is replaced by a behaviorally-adjusted dynamic index.

The internal consistency of the framework was demonstrated through retrospective case study analysis. The framework, applied to each of the three documented incidents (the McAfee EXIF exposure, the Strava heatmap aggregation, and the Advertising Identifier de-anonymization), identifies the relevant exposure surface and produces actionable remediation instructions addressing the vulnerability that was exploited in each case. As stated in the limitations (Section 2.8), this retrospective demonstration establishes coverage and internal consistency rather than predictive validity. The comparative analysis (Table 2.4) further established the framework's structural advantages over existing single-surface tools.

For practical application, it is recommended that OPRCF be deployed as an upstream classification layer that precedes the use of any individual remediation tool, enabling users to direct their remediation effort toward the highest-

impact surfaces first. The framework remains applicable to general civilian users, corporate employees, and high-risk individuals alike, ensuring that each user receives a risk index and remediation output calibrated to their selected threat context rather than a population-average assessment.

The priority direction for further work is the empirical calibration of the archetype weighting coefficients through a structured expert-elicitation study, including the construction of Analytic Hierarchy Process pairwise comparison matrices and the computation of consistency ratios. Further directions include a prospective evaluation of predictive validity on profiles with unknown outcomes; a formal sensitivity analysis quantifying the effect of each coefficient on the Final Risk Index; expansion of the rule-based prototype into an automated signal-ingestion pipeline; and extension of the archetype taxonomy to additional operational contexts.

LIST OF USED SOURCES

1. Rajamaki J., McMenamin M., Tiitta S. OSINT in Threat Intelligence [Электронный ресурс]. 2024. URL: https://www.isij.eu/system/files/download-count/2024-10/5511_OSINT_cyber_risk.pdf.
2. BRside. What Can Hackers Find About You in 5 Minutes? Your Digital Footprint Exposed [Электронный ресурс]. 2025. URL: [https://www.brside.com/blog/what-can-hackers-find-about-you-in-5-minutes-your-digital-footprint-exposed-\(2025\)](https://www.brside.com/blog/what-can-hackers-find-about-you-in-5-minutes-your-digital-footprint-exposed-(2025)).
3. Riebe T., others. Privacy Concerns and Acceptance of OSINT [Электронный ресурс]. 2023. URL: <https://petsymposium.org/popets/2023/popets-2023-0026.pdf>.
4. National Institute of Standards and Technology (NIST). Updated Privacy Framework Draft [Электронный ресурс]. 2025. URL: <https://www.maynardnexsen.com/publication-nist-releases-updated-privacyframework>.
5. Wikipedia. Open-Source Intelligence [Электронный ресурс]. 2026. URL: https://en.wikipedia.org/wiki/Open-source_intelligence.
6. ShadowDragon. What is OSINT (Open-Source Intelligence)? 2026 Guide [Электронный ресурс]. 2026. URL: <https://shadowdragon.io/blog/what-is-osint>.
7. OSINT Framework. Open-Source Intelligence Collection Methods Taxonomy [Электронный ресурс]. 2024. URL: <https://osintframework.com/>.
8. Molfar Intelligence Institute. What Is OSINT in 2025: A Guide [Электронный ресурс]. 2024. URL: <https://molfar.com/en/blog/shcho-take-osint-u-2024-gaid-vid-molfar>.
9. Barracuda Networks. Understanding the Importance of OSINT in Modern Research [Электронный ресурс]. 2025. URL: <https://blog.barracuda.com/2025/01/09/understanding-osint-modern-research>.
10. Neotas. OSINT Sources: Social Media OSINT and Investigation Techniques [Электронный ресурс]. 2025. URL: <https://www.neotas.com/osint-sources-social-media-osint/>.

11. SNS Insider. Open-Source Intelligence Market Size and Growth Report 2024 to 2032 [Электронный ресурс]. 2024. URL: <https://www.snsinsider.com/reports/open-source-intelligence-market-1390>.
12. Browne C., Abedin B., Chowdhury M. AI for OSINT Applications [Электронный ресурс]. 2024. URL: <https://link.springer.com/article/10.1007/s10207-024-00831-0>.
13. MDPI. Automated OSINT Techniques for Digital Asset Discovery and Cyber Risk Assessment [Электронный ресурс]. 2025. URL: <https://www.mdpi.com/2073-431X/14/10/430>.
14. Wilhelm A. Vice Leaves Metadata in Photo of John McAfee, Pinpointing Him to a Location in Guatemala [Электронный ресурс]. 2012. URL: <https://thenextweb.com/news/vice-leaves-metadata-in-photo-of-john-mcafee-pinpointing-him-to-a-location-in-guatemala>.
15. Global Investigative Journalism Network. Running Into Open Secrets: How to Investigate Using the Strava Fitness App [Электронный ресурс]. 2025. URL: <https://gijn.org/stories/investigations-using-strava-fitness-app/>.
16. PetaPixel. EXIF Data May Have Revealed Location of Fugitive Billionaire John McAfee [Электронный ресурс]. 2012. URL: <https://petapixel.com/2012/12/03/exif-data-may-have-revealed-location-of-fugitive-billionaire-john-mcafee/>.
17. Athena Forensics. John McAfee and the EXIF Photograph Metadata [Электронный ресурс]. 2019. URL: <https://athenaforensics.co.uk/john-mcafee-and-the-photograph-exif-metadata/>.
18. NowSecure. New NowSecure Research Targets Mobile App Privacy Risks [Электронный ресурс]. 2025. URL: <https://www.nowsecure.com/blog/2025/09/29/new-nowsecure-research-targets-mobile-app-privacy-risks-what-you-dont-see-is-hurting-you>.
19. Cybersecurity and Infrastructure Security Agency (CISA). Mobile Communications Best Practice Guidance [Электронный ресурс]. 2025. URL: https://www.cisa.gov/sites/default/files/2025-12/guidance-mobile-communications-best-practices_508c.pdf.
20. Jivoi. Awesome-OSINT: A Curated List of Amazingly Awesome OSINT [Электронный ресурс]. 2024. URL: <https://github.com/jivoi/awesome-osint>.

21. Harry C., Sivan-Sevilla I., McDermott M. Measuring the Size and Severity of the Integrated Cyber Attack Surface Across US County Governments [Электронный ресурс]. 2025. URL: <https://doi.org/10.1093/cybsec/tyae032>.
22. Rapid7 Labs. Digital Footprint Exposure: A Strategic Cyber Risk [Электронный ресурс]. 2026. URL: <https://www.rapid7.com/blog/post/tr-data-breach-digital-footprints-strategic-cyber-risk-report/>.
23. Hern A. Fitness tracking app Strava gives away location of secret US army bases [Электронный ресурс]. 2018. URL: <https://www.theguardian.com/world/2018/jan/28/fitness-tracking-app-strava-gives-away-location-of-secret-us-army-bases>.
24. Flynn J.D. The Pillar investigates: US conference gen sec resigned after use of hookup app [Электронный ресурс]. 2021. URL: <https://www.pillarcatholic.com/p/the-pillar-investigates-us-conference>.
25. TIME Magazine. A Catholic Priest's Alleged Grindr Outing and Data Privacy [Электронный ресурс]. 2021. URL: <https://time.com/6083323/bishop-pillar-grindr-data/>.
26. Frontiers in Psychology. Understanding the Privacy Disclosure Behavior and Mechanisms of Mobile Users [Электронный ресурс]. 2025. URL: <https://www.frontiersin.org/journals/psychology/articles/10.3389/fpsyg.2025.1535533/full>.
27. Nature Scientific Reports. An Exploration of the Influencing Factors of Privacy Fatigue Among Mobile Users [Электронный ресурс]. 2025. URL: <https://www.nature.com/articles/s41598-024-84646-z>.
28. New America Foundation. Preserving Privacy: An Impact Framework for OSINT [Электронный ресурс]. 2025. URL: <https://www.newamerica.org/insights/preserving-privacy-an-impact-framework/>.
29. New America Foundation. Explaining and Developing the OSINT Privacy Impact Framework (OPIF) [Электронный ресурс]. 2026. URL: <https://www.newamerica.org/insights/preserving-privacy-an-impact-framework/explaining-and-developing-the-osint-privacy-impact-framework-opif/>.

30. University of Texas Center for Identity. Privacy Risk Predictions Based on Fundamental Understanding of Personal Data and an Evolving Threat Landscape [Электронный ресурс]. 2025. URL: <https://arxiv.org/abs/2508.04542>.
31. Greitzer F.L., Hohimer R.E. Modeling Human Behavior to Anticipate Insider Attacks [Электронный ресурс]. 2011. URL: <https://digitalcommons.usf.edu/jss/vol4/iss2/3/>.
32. Yilmaz I., Can O. AI for Insider Threat Detection [Электронный ресурс]. 2024. URL: <https://www.etasr.com/index.php/ETASR/article/view/6911>.
33. Springer. Human Factors in Cybersecurity: An Interdisciplinary Review [Электронный ресурс]. 2025. URL: <https://link.springer.com/article/10.1007/s10207-025-01032-0>.
34. WeLiveSecurity. The OSINT Playbook: Find Your Weak Spots Before Attackers Do [Электронный ресурс]. 2025. URL: <https://www.welivesecurity.com/en/privacy/osint-playbook-find-weak-spots-attackers-do>.
35. Moreira F.R., others. Cybersecurity Risk Assessment Through Analytic Hierarchy Process: Integrating Multicriteria and Sensitivity Analysis [Электронный ресурс]. 2025. URL: <https://www.scitepress.org/Papers/2025/131973/131973.pdf>.
36. Hevner A.R., others. Design Science in Information Systems Research [Электронный ресурс]. 2004. URL: <https://www.jstor.org/stable/25148625>.
37. Montjoye Y.-A. de и др. Unique in the Crowd: The Privacy Bounds of Human Mobility [Электронный ресурс]. 2013. URL: <https://www.nature.com/articles/srep01376>.
38. Sweeney L. Simple Demographics Often Identify People Uniquely [Электронный ресурс]. 2000. URL: <https://dataprivacylab.org/projects/identifiability/paper1.pdf>.
39. Narayanan A., Shmatikov V. Robust De-Anonymization of Large Sparse Datasets [Электронный ресурс]. 2008. URL: https://www.cs.utexas.edu/~shmat/shmat_oak08netflix.pdf.

40. Electronic Frontier Foundation (EFF). Surveillance Self-Defense: Privacy Breakdown of Mobile Phones [Электронный ресурс]. 2025. URL: <https://ssd.eff.org/playlist/privacy-breakdown-of-mobile-phones>.
41. Electronic Frontier Foundation (EFF). How to: Get to Know Android Privacy and Security Settings [Электронный ресурс]. 2025. URL: <https://ssd.eff.org/module/how-to-get-to-know-android-privacy-and-security-settings>.